

Smart Metro Ticket Booking System

REQUIREMENT ANALYSIS PHASE

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks

8. Technology Stack

8.1 Platform Overview

The Smart Metro Ticket Booking System is built natively on the ServiceNow platform, using its low-code application development, workflow automation, and integration capabilities. This keeps the solution within a single governed platform, reducing integration overhead while still supporting external payment and notification services.

8.2 Layered Architecture

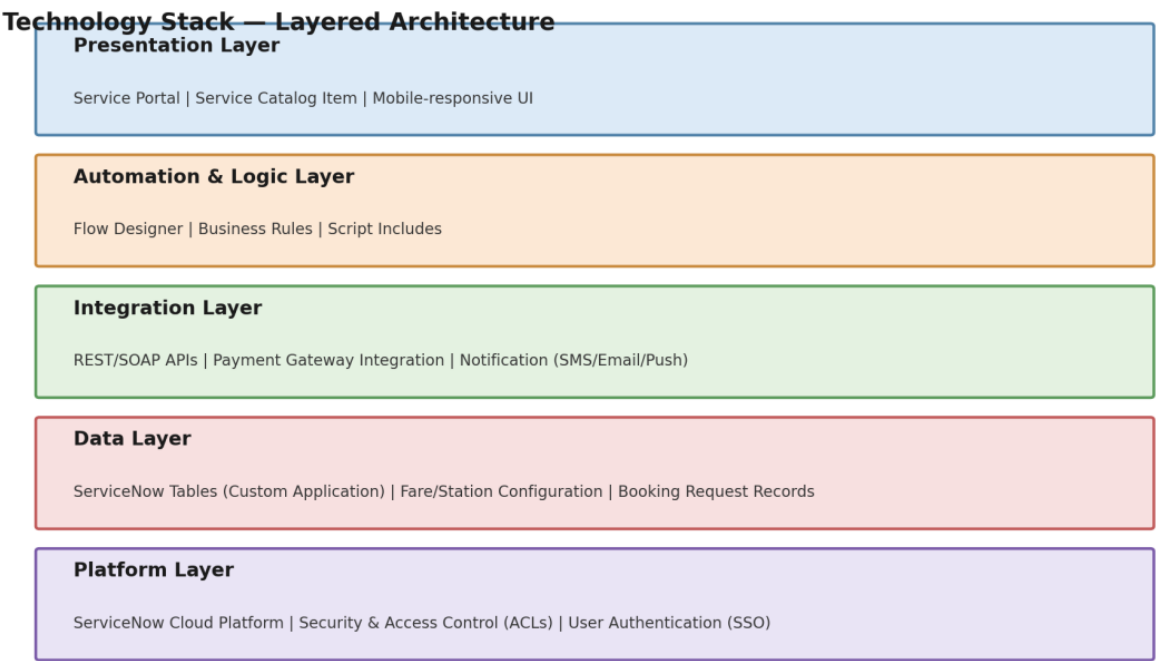


Figure 8.1: Layered technology stack of the Smart Metro Ticket Booking System

8.3 Core ServiceNow Components

Component	Role in the Solution
Service Portal / Service Catalog	Provides the commuter-facing booking form for station, date, and passenger selection

Component	Role in the Solution
Flow Designer	Automates fare calculation, ticket generation, and notification triggering
Business Rules / Script Includes	Enforces validation logic (e.g. valid station pairs, passenger limits)
Custom Tables (App Engine)	Stores fare rules, station master data, booking requests, and ticket records
Notify / Notification Engine	Delivers booking confirmation and digital ticket via SMS/Email/Push
Access Control Lists (ACLs)	Restricts data visibility so commuters see only their own bookings
Performance Analytics / Reporting	Powers the admin dashboard for bookings, revenue, and traffic (future phase)

8.4 External Integrations

Integration	Technology	Interface	Purpose
Payment Gateway	UPI / Card / Wallet	REST API	Processes fare payment and returns transaction status
QR Code Generation	QR/barcode generation library or REST-based service	REST API / Script	Generates the scannable digital ticket
SMS Gateway	Third-party SMS provider	REST API	Delivers ticket and confirmation as SMS
Email Service	SMTP / Email provider	SMTP / REST API	Delivers ticket and confirmation as Email

8.5 Security & Access

- User authentication via ServiceNow instance login / Single Sign-On (SSO).
- Role-based Access Control Lists (ACLs) to separate commuter, admin, and system access.
- No card or sensitive payment data stored on the ServiceNow instance; handled entirely by the PCI-compliant payment gateway.
- All booking, payment, and cancellation actions logged for audit and traceability.

8.6 Justification

Building on native ServiceNow components (Service Catalog, Flow Designer, and Notifications) allows the Phase 1 MVP to be delivered quickly with minimal custom code, while still leaving clean integration points for payment gateways and external notification channels. This approach directly supports the P0 (Must Have) requirements identified in the Solution Requirements document and keeps the architecture extensible for P1–P3 features in later phases.